

Affine orthogonal quiver varieties at δ and 2δ , and their contact links

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Abstract

Pinot recently computed the quiver varieties $\mathfrak{M}_\delta$ and $\mathfrak{M}_{2\delta}$ of the orthogonal quivers whose underlying graph is an affine Dynkin diagram, and identified Kleinian surface singularities among them [14]. We make three observations and one construction. (i) Using only the closed immersion of Li [9, Prop. 9.2.1], whenever $\mathfrak{M}_\delta$ is a surface the involution acts trivially and $\mathfrak{M}_\delta$ is the classical Kleinian surface of the underlying affine diagram. (ii) Consequently the two $\tilde{D}$ rows of [14, Thm. 1.2] are of type D_{2n-2} and D_{2n-1} , not D_{n+1} ; for $(\tilde{D}_{2n-1}, a)$ the defining relation is $z^2 = z(-x)^{n-1} - xy^2$. We confirm both by an independent numerical computation of the invariant ring. (iii) In every row at 2δ the group Γ contains the classical group of the underlying graph with index 2 (or 1); we conjecture that the 2δ surfaces are quotients of the classical surface by an involution, and prove the corresponding statement for the coordinate rings. Finally, each surface is a cone whose link is a contact 3-manifold S^3/Γ ; we show that the symmetric representation space is quaternionic, so the link is the 3-Sasakian (Kempf-Ness) quotient of the unit sphere, and we tabulate the Reeb (Hopf) Seifert fibrations.

How to read this version. Every statement carries one tag: [proved here], [cited], [numerically checked] (a script in the bundle; evidence, not proof), [kernel-checked arithmetic] (Lean 4, arithmetic only), or [open]. **Verification bundle** (Python and Lean, with a README): [click here to open the attached zip](https://totogt.github.io/geometry/collab/pinot-orthogonal-quivers/) (attached to this PDF), or download it from <https://totogt.github.io/geometry/collab/pinot-orthogonal-quivers/>. This text was prepared with the assistance of Claude (Anthropic); the author takes responsibility for it, and it has not yet been checked by V. Pinot.

1 Introduction

Nakajima quiver varieties of affine Dynkin type recover Kronheimer’s ALE spaces and the Kleinian singularities $\mathbb{C}^2/\Gamma$ [7, 12]; the ADE labels go back to Gabriel’s classification of quivers of finite type [5]. Pinot studies the analogous quotients for *orthogonal quivers* [14]: quivers Q with a contravariant involution τ and a sign s , whose symmetric representations $\text{SRep}(\overline{Q}, d)$ carry an action of $G_d^b = \prod O(d_i) \times \prod GL(d_j)$, and

$$\mathfrak{M}_d = \mu^{-1}(0) // G_d^b.$$

His Theorem 1.2 lists $\mathfrak{M}_\delta$ and $\mathfrak{M}_{2\delta}$ for ten infinite families. This note re-reads that table from two angles: the comparison with the classical quiver variety of the same graph (§§3–4), and the contact geometry of the link of each surface (§5).

Results.

- **Theorem A** (§3). If $\mathfrak{M}_\delta$ is a surface, then $\mathfrak{M}_\delta \cong \mathfrak{M}_\delta^{\text{cl}}$, the classical Kleinian surface of the underlying affine graph, and the involution acts trivially on it. Conditional only on the applicability of Li's closed immersion to Pinot's unframed setting.
- **Corollary B** (§3). The $\tilde{D}$ rows read D_{2n-2} and D_{2n-1} . The $(\tilde{D}_{2n-2}, v)$ equation of [14] is correct; the $(\tilde{D}_{2n-1}, a)$ relation is $z^2 = z(-x)^{n-1} - xy^2$. Confirmed numerically for $n = 3, 4, 5$ by computing the invariant ring directly.
- **Conjecture C** (§4). Each 2δ surface is $\mathfrak{M}_\delta^{\text{cl}}/\iota$ for an involution ι ; equivalently $\Gamma_{2\delta} \supset \Gamma_{\text{cl}}$ with index ≤ 2 , and the classical link double-covers the 2δ link. The ring-level statement is proved.
- **Theorem D** (§5). $\text{SRep}(\overline{Q}, d)$ is a quaternionic subspace; the link is the Kempf-Ness / 3-Sasakian quotient of the unit sphere; the Reeb flow is the Hopf flow, with the Seifert data of Table 2.

2 Setting

We use the notation of [14] throughout, with the standard signed space ($J_i = \text{id}$ in suitable bases), so that the involution on representations is $\sigma(v)_a = s(a) v_{\tau a}^T$ and $\text{SRep} = \text{Rep}^\sigma$ [14, Def. 2.3, 2.5]. The moment map is the restriction of the quiver moment map, $\mu(\text{SRep}) \subset \mathfrak{g}^b$ [14, Cor. 2.13]. The invariant ring is generated by traces of cycles [14, Thm. 3.9], [8]. The scaling $t \cdot v$ gives $\text{tr}(\Lambda^k C)$ the weight $k \ell(C)$, so every $\mathfrak{M}_d$ is a cone [14, §3.5.1].

The comparison map. Let $\mathfrak{M}_d^{\text{cl}} = \mu^{-1}(0)//G_d$ be the classical quiver variety of the same quiver. By universality of categorical quotients there is a morphism $\iota : \mathfrak{M}_d \rightarrow (\mathfrak{M}_d^{\text{cl}})^{[\sigma]}$. Remark 2.15 of [14] states that it is an isomorphism, citing [9, 13]. The sources give less: Li proves that ι is a *closed immersion* for an arbitrary graph [9, Prop. 9.2.1], calls the isomorphism conjectural, and proves it in type A_n with alternating signs [9, Prop. 9.2.2]; Nakajima treats only smooth σ -quiver varieties (generic ζ) [13]. Li works with framed varieties (v, w) and prescribed forms; we *assume* throughout that his Proposition 9.2.1 applies to Pinot's unframed signed setting (Question 2). We use only the closed immersion.

3 The surfaces at δ

Theorem 3.1. *Assume Li's closed immersion $\iota : \mathfrak{M}_\delta \hookrightarrow \mathfrak{M}_\delta^{\text{cl}}$ holds in Pinot's setting, δ being the minimal imaginary root of the underlying affine graph. If $\mathfrak{M}_\delta$ is a reduced irreducible surface (as in all non-trivial rows of [14, Thm. 1.2]), then ι is an isomorphism and $[\sigma]$ acts trivially on $\mathfrak{M}_\delta^{\text{cl}}$. In particular $\mathfrak{M}_\delta \cong \mathbb{C}^2/\Gamma$ with Γ the McKay group of the underlying affine graph.*

Proof. $\mathfrak{M}_\delta^{\text{cl}} \cong \mathbb{C}^2/\Gamma$ is an irreducible surface [7, 12]. A closed immersion of a reduced irreducible surface into an irreducible surface is surjective, hence an isomorphism onto it. Since its image lies in the fixed locus of $[\sigma]$, that fixed locus is everything. [proved here] (conditional on the hypothesis). $\square$

Corollary 3.2. *The non-trivial δ rows are of type A_{2n-2} for $\tilde{A}_{2n-2}$, A_{2n-1} for $\tilde{A}_{2n-1}$, D_{2n-2} for $(\tilde{D}_{2n-2}, v)$ and D_{2n-1} for $(\tilde{D}_{2n-1}, a)$. The last two replace the label D_{n+1} printed in [14, Thm. 1.2]; they agree with it only for $n = 3$ in the first case, and never in the second.*

Corollary 3.2 depends on the hypothesis of Theorem 3.1. The next two results check it independently.

Proposition 3.3 (weighted monomials). *In the $\tilde{D}$ rows let $x = \det C_3$, $y = \text{tr}(C_1 C_2)$, $z = \text{tr}(C_1 C_2 C_3)$, with $L = 2n - 5$ for $(\tilde{D}_{2n-2}, v)$ and $L = 2n - 4$ for $(\tilde{D}_{2n-1}, a)$ [14, p. 18]. Their weights are $(4, 2L + 2, 2L + 4)$, and a relation homogeneous with z^2 has degree $4L + 8$. The only monomials of that degree are:*

- $(\tilde{D}_{2n-2}, v)$, $n \geq 4$: $z^2, xy^2, x^{n-1}y, x^{2n-3}$. The relation $z^2 + \beta xy^2 + \alpha x^{n-1}y + \gamma x^{2n-3}$ is of type D_{2n-2} if and only if $\beta \neq 0$ and $\gamma \neq \alpha^2/4\beta$. For $n = 3$ the weights are $(4, 4, 6)$, so y^3 also has degree 12: the relation is $z^2 + c(x, y)$ with c a binary cubic, of type D_4 if and only if c has three distinct linear factors.
- $(\tilde{D}_{2n-1}, a)$: $z^2, x^{n-1}z, xy^2, x^{2n-2}$. The relation $z^2 + \alpha x^{n-1}z + \beta xy^2 + \gamma x^{2n-2}$ is of type D_{2n-1} if and only if $\beta \neq 0$ and $\gamma \neq \alpha^2/4$. The term $x^{n-1}y$ printed in [14, Lemma 3.27] has degree $4L + 6$ and cannot occur.

Proof. Solve $4a + (2L + 2)b + (2L + 4)c = 4L + 8$ in non-negative integers. Then complete the square: in y for the first row, in z for the second. The result is the normal form $z^2 + xy^2 + cx^{m-1}$ of D_m with $m = 2n - 2$, resp. $m = 2n - 1$, and $c \neq 0$ exactly under the stated conditions. Equivalently, by the Milnor-Orlik formula $\mu = \prod (d/w_i - 1) = L + 3$ [11]. [proved here] [kernel-checked arithmetic] $\square$

Proposition 3.4 (direct computation). *For $n = 3, 4, 5$ we sampled 80 random points of $\mu^{-1}(0) \cap \text{SRep}(\overline{Q}, \delta)$ by Gauss-Newton and found all linear relations among the monomials of weight $\leq 4L + 8$ in x, y, z . In each case there is exactly one relation (relative singular value $\leq 2 \cdot 10^{-8}$, against ≥ 0.1 for the next), with integer coefficients:*

$$(\tilde{D}_{2n-2}, v) : z^2 = -y(-x)^{n-1} - xy^2, \quad (\tilde{D}_{2n-1}, a) : z^2 = z(-x)^{n-1} - xy^2.$$

The first is the equation of [14, Lemma 3.28]. The second is [14, Lemma 3.27] with y replaced by z in the first term: the step $\text{tr}(C_1 C_2^2 C_3^3) = \text{tr}(C_1 C_2 C_3)(-\det C_3)^{n-1}$ appears to have been written with $\text{tr}(C_1 C_2)$. In both rows the non-degeneracy conditions of Proposition 3.3 hold ($\beta = 1$, discriminant $-1/4$; for $n = 3$ in the first row the cubic is $xy(x + y)$, with three distinct factors). For sign -1 in $(\tilde{D}_{2n-1}, a)$, all invariants vanish, in agreement with [14]. [numerically checked]

4 The surfaces at 2δ

At 2δ the classical variety is 4-dimensional, and the orthogonal surfaces are new. Comparing each $\Gamma_{2\delta}$ with the classical group Γ_{cl} of the same graph:

2δ row	$\Gamma_{2\delta}$	Γ_{cl}	index
$(\tilde{A}_{2n-2}, v-a), -1$	$\mathbb{Z}_{4n-2}$	$\mathbb{Z}_{2n-1}$	2
$(\tilde{A}_{2n-1}, a-a), (+, -)$	$\mathbb{Z}_{4n}$	$\mathbb{Z}_{2n}$	2
$(\tilde{A}_{2n-1}, c)$	BD_{4n}	$\mathbb{Z}_{2n}$	2
$(\tilde{A}_{2n-1}, a-a), (-, -)$	$\mathbb{Z}_{2n}$	$\mathbb{Z}_{2n}$	1

quiver	sign	d	equation	type	Γ	$ \Gamma $
$(\tilde{A}_{2n-2}, v-a)$	+1	δ	$xy = z^{2n-1}$	A_{2n-2}	$\mathbb{Z}_{2n-1}$	$2n-1$
	-1	2δ	$xy = z^{4n-2}$	A_{4n-3}	$\mathbb{Z}_{4n-2}$	$4n-2$
$(\tilde{A}_{2n-1}, a-a)$	(+, +)	δ	$xy = z^{2n}$	A_{2n-1}	$\mathbb{Z}_{2n}$	$2n$
	(+, -)	2δ	$xy = z^{4n}$	A_{4n-1}	$\mathbb{Z}_{4n}$	$4n$
	(-, -)	2δ	$xy = z^{2n}$	A_{2n-1}	$\mathbb{Z}_{2n}$	$2n$
$(\tilde{A}_{2n-1}, v-v)$		δ	$xy = z^{2n}$	A_{2n-1}	$\mathbb{Z}_{2n}$	$2n$
$(\tilde{A}_{2n-1}, c)$		2δ	$x^{n+1} - y^2x - z^2 = 0$	D_{n+2}	BD_{4n}	$4n$
$(\tilde{D}_{2n-2}, v)$		δ	$z^2 = -y(-x)^{n-1} - xy^2$	D_{2n-2}	$\text{BD}_{8(n-2)}$	$8n-16$
$(\tilde{D}_{2n-1}, a)$	+1	δ	$\mathbf{z}^2 = \mathbf{z}(-\mathbf{x})^{n-1} - \mathbf{xy}^2$	D_{2n-1}	$\text{BD}_{4(2n-3)}$	$8n-12$

Table 1: The non-trivial rows of [14, Thm. 1.2], with the corrections in bold. BD_{4k} is the binary dihedral group of order $4k$ (for $n = 1$ in the $(\tilde{A}_{2n-1}, c)$ row, $D_3 = A_3$). All other rows are reduced points.

The two index-2 overgroups of $\mathbb{Z}_{2n}$ in $SU(2)$ that occur here, $\mathbb{Z}_{4n}$ and BD_{4n} , both appear.

Conjecture 4.1. Each 2δ surface is isomorphic to $\mathfrak{M}_\delta^{\text{cl}}/\iota$ for an involution ι of the classical surface, trivial exactly in the $(-, -)$ row.

Proposition 4.2. Write $\mathbb{C}^2/\mathbb{Z}_m = \{XY = Z^m\}$ with $X = u^m, Y = v^m, Z = uv$.

- (a) The generator of $\mathbb{Z}_{2m} \supset \mathbb{Z}_m$ induces $\iota_1 : (X, Y, Z) \mapsto (-X, -Y, Z)$. Its invariants X^2, Y^2, Z satisfy $X^2Y^2 = Z^{2m}$, the surface A_{2m-1} .
- (b) For $m = 2n, j \in \text{BD}_{4n}$ induces $\iota_2 : (X, Y, Z) \mapsto (Y, X, -Z)$. Its invariants $s = X + Y, w = Z^2, t = Z(X - Y)$ satisfy $t^2 = ws^2 - 4w^{n+1}$, the surface D_{n+2} .

With $m = 2n - 1$, resp. $m = 2n$, these reproduce the three index-2 rows above exactly.

Proof. Direct substitution; the invariant rings are the classical Klein invariants of $\mathbb{Z}_{2m}$ and BD_{4n} . [proved here] [numerically checked] (symbolic, $m \leq 11$). $\square$

Pinot's own generators fit this picture. In the $(+, -)$ row the 2δ coordinates are $\det C_1, \det C_2$ and $\text{tr } C_3$, where the classical δ coordinates are $\text{tr } C_1, \text{tr } C_2, \text{tr } C_3$. The weights double for the first two, just as $X \mapsto X^2, Y \mapsto Y^2$ in (a). The remaining step is to construct ι from the quiver data. [open]

5 The contact links

5.1 Cones and weights

Lemma 5.1. In every row of Table 1, Pinot's path-length weights of the three generators equal the degrees of the corresponding Klein invariants in $\mathbb{C}[u, v]^\Gamma$: $(m, m, 2)$ for $\mathbb{Z}_m$ and $(4, 2k, 2k + 2)$ for BD_{4k} . Hence the identification $\mathfrak{M}_d \cong \mathbb{C}^2/\Gamma$ can be chosen graded, and the unit circle of the scaling $\mathbb{C}^*$ acts on $\mathbb{C}^2/\Gamma$ by scalars.

Proof. At δ for $\tilde{A}_{m-1}$: $\text{tr } C_1, \text{tr } C_2$ have weight $m, \text{tr } C_3$ weight 2. At 2δ , determinants double the cycle length: $(4n - 2, 4n - 2, 2)$ and $(4n, 4n, 2)$. In the $(\tilde{A}_{2n-1}, c)$ row the weights are $(4, 2n, 2n + 2)$, and in the $\tilde{D}$ rows $(4, 2L + 2, 2L + 4)$ with $k = L + 1$. The normal-form changes of variables used above preserve the grading. [proved here] $\square$

The link $L_Q = (\mathfrak{M}_d \setminus 0)/\mathbb{R}_{>0} \cong S^3/\Gamma$ carries its canonical contact structure: the complex tangencies, unique up to contactomorphism [4], which is the quotient of the standard structure of S^3 since $\Gamma \subset SU(2)$. It is universally tight and Milnor fillable. [cited]

5.2 A real structure

A real contact form needs a real exact symplectic form. Pinot's ω and μ are holomorphic, so we use the flat hyperkähler structure of $T^*\text{Rep}(Q, d)$: $I = i$ and $J(x, y) = (-y^\dagger, x^\dagger)$ arrow by arrow.

Lemma 5.2. *In the standard signed space, $\sigma \circ J = J \circ \sigma$ and σ is unitary. Hence $\text{SRep}(\overline{Q}, d)$ is a quaternionic subspace. The compact group $K^b = G^b \cap \prod U(d_i) = \prod O(d_i, \mathbb{R}) \times \prod U(d_j)$ preserves I, J, K , and both moment maps send SRep into $\mathfrak{k}^b$.*

Proof. $\sigma(Jv)_a = s(a)(-v_{(\tau a)^*}^\dagger)^T = -s(a)\overline{v_{(\tau a)^*}} = J(\sigma v)_a$, and similarly for a^* , using $s(a^*) = s(a)$ and $\tau(a^*) = \tau(a)^*$. σ permutes matrix entries up to sign and transposition, so it preserves the Hermitian norm. For $\mu_{\mathbb{C}}$ the last claim is [14, Prop. 2.11]; for $\mu_{\mathbb{R}}$ the same computation applies. [proved here] [numerically checked] $\square$

Theorem 5.3. *There is a $\mathbb{C}^*$ -equivariant homeomorphism $(\mu_{\mathbb{R}}^{-1}(0) \cap \mu_{\mathbb{C}}^{-1}(0) \cap \text{SRep})/K^b \cong \mathfrak{M}_d$. Intersecting with the unit sphere gives*

$$L_Q \cong (\mu_{\mathbb{R}}^{-1}(0) \cap \mu_{\mathbb{C}}^{-1}(0) \cap S(\text{SRep}))/K^b,$$

the 3-Sasakian quotient of the unit sphere in the quaternionic space SRep [2, 1]. The smooth structure of L_Q comes from Lemma 5.1. The reduction need not be locally free on the zero set.

Proof. $\mu_{\mathbb{C}}^{-1}(0) \cap \text{SRep}$ is a closed G^b -stable subvariety of a unitary representation of $G^b = (K^b)_{\mathbb{C}}$. Kempf-Ness [6] identifies its categorical quotient with the K^b -quotient of the zero set of the K^b -moment map, which is $\mu_{\mathbb{R}}$ by Lemma 5.2. Both sides are cones, so the identification restricts to links. [proved here] (standard) $\square$

5.3 Reeb flow

By Lemma 5.1, the Reeb flow of the contact form induced by the flat metric is the Hopf flow $x \mapsto e^{i\theta}x$ on S^3/Γ .

Proposition 5.4. *The Hopf flow on S^3/Γ is a Seifert fibration with the following data. The generic orbit has period 2π if $-1 \notin \Gamma$ and π otherwise. The exceptional fibre over a cone point of order a has period (generic)/ a .*

Proof. The stabiliser of x is $\{\theta : e^{i\theta}x \in \Gamma x\}$. For generic x only the scalars $\pm 1 \in \Gamma$ contribute. The base is $\mathbb{C}P^1/\bar{\Gamma}$, with $\bar{\Gamma} \subset SO(3)$ cyclic of order m or $m/2$, or dihedral of order $2k$. Its cone points are the points of $\mathbb{C}P^1$ with non-trivial stabiliser in $\bar{\Gamma}$. The identity $\chi^{\text{orb}} = 2/|\bar{\Gamma}|$ checks the data. [proved here] [numerically checked] [kernel-checked arithmetic] $\square$

Under Conjecture 4.1, $S^3/\Gamma_{\text{cl}} \rightarrow S^3/\Gamma_{2\delta}$ is a double covering of contact manifolds commuting with the Reeb flows: the link at 2δ is a $\mathbb{Z}_2$ -quotient of the classical link.

type	Γ	L_Q	base orbifold	generic period
$A_{m-1}, m \text{ odd}$	$\mathbb{Z}_m$	$L(m, m-1)$	$S^2(m, m)$	2π
$A_{m-1}, m \text{ even}$	$\mathbb{Z}_m$	$L(m, m-1)$	$S^2(m/2, m/2)$	π
D_{k+2}	BD_{4k}	prism manifold	$S^2(2, 2, k)$	π

Table 2: Seifert data of the Reeb flow: base orbifold and multiplicities. The Seifert invariants β_i and the Euler number are not computed here.

5.4 Fillings

The classical minimal resolution of $\mathbb{C}^2/\Gamma$ (Kronheimer’s $\zeta \neq 0$ quotient) is a strong symplectic (Kähler) filling of L_Q . It is not Stein, since it contains the exceptional (-2) -curves. For ADE singularities it is diffeomorphic to the Milnor fibre [3], which is a Stein filling. For lens spaces with this contact structure all fillings are classified by Lisca [10]. Whether Pinot’s orthogonal quotient at $\zeta \neq 0$ gives a resolution is open: only the GL factors of G^b admit characters, and the involution sends ζ to $-\zeta$ before a Weyl correction [13]. [cited] [open]

6 Questions for V. Pinot

Question 1. Do you confirm the labels D_{2n-2}, D_{2n-1} and the relation $z^2 = z(-x)^{n-1} - xy^2$ of Corollary 3.2 and Proposition 3.4?

Question 2. Does Li’s Proposition 9.2.1 apply to your unframed signed setting (for instance with $w = 0$, or with an auxiliary framing vertex)? Theorem 3.1 uses only this closed immersion.

Question 3. Do you have a source for the isomorphism of Remark 2.15 at $\zeta = 0$ beyond type A_n with alternating signs?

Question 4. Do the 2δ surfaces arise as quotients of the classical surface by an involution (Conjecture 4.1)? Is there a natural candidate for ι in terms of the quiver?

Question 5. Do you plan to treat the case of your Remark 1.3, $(\tilde{D}_{2n-1}, a)$ with sign -1 at 2δ ? Under Conjecture 4.1 one would expect an overgroup of $\text{BD}_{4(2n-3)}$ of index at most 2.

A Verification bundle

[Attached zip](#); also at <https://totogt.github.io/geometry/collab/pinot-orthogonal-quivers/>. Python 3 with numpy, scipy, sympy; Lean 4 core (no Mathlib).

- `verify_Dtilde_relation_numeric.py`: Proposition 3.4; saved output included.
- `verify_ade_types.py`: Milnor numbers of the $\tilde{D}$ -row polynomials; homogeneity defect of the printed relation.
- `verify_2delta_index2.py`: Proposition 4.2.
- `verify_quaternionic_SRep.py`: Lemma 5.2.

- `verify_seifert.py`: Table 2 and the group orders.
- `DM3Quivers_Check.lean`: weight degrees, the Milnor-Orlik numerator identity $(d-w_x)(d-w_y)(d-w_z) = (L+3)w_xw_yw_z$, group orders, and the orbifold identities. It does not check Remark 2.15 or any geometric statement.

References

- [1] C. P. Boyer, K. Galicki, *3-Sasakian manifolds*, Surveys in Differential Geometry VI (1999); arXiv:hep-th/9810250.
- [2] C. P. Boyer, K. Galicki, B. M. Mann, *The geometry and topology of 3-Sasakian manifolds*, J. Reine Angew. Math. 455 (1994), 183-220.
- [3] E. Brieskorn, *Singular elements of semi-simple algebraic groups*, Actes du Congrès International des Mathématiciens (Nice, 1970), Tome 2, 279-284.
- [4] C. Caubel, A. Némethi, P. Popescu-Pampu, *Milnor open books and Milnor fillable contact 3-manifolds*, Topology 45 (2006), 673-689.
- [5] P. Gabriel, *Unzerlegbare Darstellungen I*, Manuscripta Math. 6 (1972), 71-103.
- [6] G. Kempf, L. Ness, *The length of vectors in representation spaces*, Algebraic Geometry (Copenhagen 1978), Lecture Notes in Math. 732, Springer, 1979, 233-243.
- [7] P. B. Kronheimer, *The construction of ALE spaces as hyper-Kähler quotients*, J. Differential Geom. 29 (1989), 665-683.
- [8] L. Le Bruyn, C. Procesi, *Semisimple representations of quivers*, Trans. Amer. Math. Soc. 317 (1990), 585-598.
- [9] Y. Li, *Quiver varieties and symmetric pairs*, Represent. Theory 23 (2019), 1-56.
- [10] P. Lisca, *On symplectic fillings of lens spaces*, Trans. Amer. Math. Soc. 360 (2008), 765-799.
- [11] J. Milnor, P. Orlik, *Isolated singularities defined by weighted homogeneous polynomials*, Topology 9 (1970), 385-393.
- [12] H. Nakajima, *Instantons on ALE spaces, quiver varieties, and Kac-Moody algebras*, Duke Math. J. 76 (1994), 365-416.
- [13] H. Nakajima, *Instantons on ALE spaces for classical groups, involutions on quiver varieties, and quantum symmetric pairs*, arXiv:2510.13007 (2025).
- [14] V. Pinot, *Quiver varieties for affine orthogonal quivers*, arXiv:2609.39434 (2026).